

Quiz 01. Classifying ODEs
prepared by C. Cahig for ENS181 (AY 2026-2027, S1)

Scenario. For each equation, identify if it is variable-separable, homogeneous¹, exact², linear³, or none. An equation may belong to multiple categories at once (say, linear and homogeneous and separable). In all cases, the character d is reserved for derivatives and differentials so that, for instance, $\frac{d^*}{d\omega}$ reads “the derivative of $\star$ with respect to ω ” and $d\alpha$ is “the differential of α ”. Following convention, e is the base of the natural logarithm, and $\exp(abc) = e^{abc}$.

Item 01.

$$e^\beta \frac{d\alpha}{d\beta} = e^\beta$$

Item 02.

$$\left(\frac{u}{v}\right)^3 \frac{du}{dv} + 1 = 0$$

Item 03.

$$\frac{R}{S} \frac{dS}{dR} - 1 = 0$$

Item 04. Assume $\cos w \neq 0$.

$$\cos w \dot{w} - t \cos w = 0$$

Item 05.

$$\phi \frac{d\theta}{d\phi} + \theta = 2\phi$$

Item 06.

$$m^2 \frac{dn}{dm} = n^2$$

Item 07.

$$2p \frac{dp}{dq} + 3q^2 = 0$$

Item 08.

$$\frac{1}{K} \frac{dK}{dT} - T = 0$$

Item 09.

$$2uv \frac{du}{dv} + u^2 = v^2$$

Item 10.

$$B \frac{dA}{dB} = A - B$$

Item 11.

$$s^2 \frac{dr}{ds} + 2rs = -\cos s$$

Item 12.

$$\frac{dw}{dz} - e^w \sin z = 0$$

Item 13.

$$p^2 \frac{dq}{dp} + q^2 = pq$$

Item 14.

$$2\lambda\omega \frac{d\lambda}{d\omega} + \lambda^2 + \cos \omega = 0$$

Item 15. Assume $\cos \Psi \neq 0$.

$$\cos \Psi \frac{d\Gamma}{d\Psi} + \Gamma \sin \Psi = \Psi^2 \cos \Psi$$

Item 16.

$$\frac{d\eta}{d\xi} - \eta - \xi^2 = 1$$

Item 17.

$$\exp\left(\frac{dp}{dq}\right) = p + q$$

Item 18.

$$c^3 \frac{d\Delta}{dc} + 3c^2 \Delta = 0$$

Item 19.

$$(e^b + be^a) \frac{da}{db} + ae^b + e^a = 0$$

Item 20.

$$(J^2 + 1) \frac{dJ}{dK} - K^2 = 0$$

Marking. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.

¹ Homogeneity here is not according to linear ODEs whose standard forms have zero right-hand sides. Instead, think of homogeneity of functions. See [Sections 2.2 and 2.3 of Supplemental Material 01](#).

² Test only the given equations for exactness. Do not look for an integrating factor.

³ Recall the process of rewriting the ODE into such that derivatives of the dependent variable is on the left-hand side, examining if the right-hand side is a constant or a function of the independent variable alone, and examining if each of the coefficients of the derivatives is a constant or a function of the independent variable alone. Refer to [Section 1.1](#) of AEM05.